

AI Tool Usage Disclosure

Paper: Hierarchical Deep Learning for Carbonate Grain Classification in Thin-Section Images **Venue:** UBMK 2026

AI-based tools were used only as language-editing and coding aids during the preparation of this paper. All research ideas, study design, methodology, experiments, analysis, and interpretation are the authors' own work. The authors reviewed all tool output and take full responsibility for the content of the paper.

Which tool was used in which part

AI tool	Where used	How it was used
Anthropic Claude	Experiment code; manuscript text (all sections)	Assistance with software/code for the experiments; grammar and sentence-level language editing
OpenAI Codex	Manuscript text (all sections); LaTeX source	Grammar and sentence-level language editing; checking LaTeX consistency

AI tools were **not** used to generate scientific content, experimental results, data, tables, or figures.